

Диплом
Кононов Александр Михайлович
18.02.2026

Задача:

Условие:

Решение:

$$\hat{H} = g \left(\hat{b}_{-}^{\dagger} \hat{a}_{+} + \hat{b}_{-} \hat{a}_{+}^{\dagger} \right)$$

$$|\psi_{in}\rangle = e^{-|\alpha|^2/2} e^{\alpha \hat{b}_H^{\dagger}} |0\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n_H\rangle$$

$$|\psi_{in}^H\rangle = e^{\alpha \hat{b}_H^{\dagger}} |0\rangle = \sum_n \frac{\alpha^n}{\sqrt{n!}} |n_H\rangle$$

$$|\psi_{out}\rangle = e^{-i\hat{H}t} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n_H\rangle = e^{\alpha \hat{b}_{+}^{\dagger}/\sqrt{2}} \hat{S}_0 e^{\alpha \hat{b}_{-}^{\dagger}/\sqrt{2}} |0\rangle$$

$$|\psi_{out}\rangle = \frac{e^{-|\alpha|^2/2} e^{\alpha \hat{b}_H^{\dagger}}}{2} \left[e^{-i\theta/2} e^{-i\gamma \hat{b}_{-}^{\dagger}} + e^{i\theta/2} e^{i\gamma \hat{b}_{-}^{\dagger}} + \right. \\ \left. + \left(e^{-i\theta/2} e^{-i\gamma \hat{b}_{-}^{\dagger}} - e^{i\theta/2} e^{i\gamma \hat{b}_{-}^{\dagger}} \right) \hat{a}_{+}^{\dagger} \right] |0\rangle$$

$$|\psi_{out}^V\rangle \approx \cos\left(\frac{\theta}{2}\right) |0\rangle + \frac{i\delta - 1}{\sqrt{2}} \sin\left(\frac{\theta}{2}\right) |1\rangle + \frac{i\delta}{2} \cos\left(\frac{\theta}{2}\right) |2\rangle$$

$$|\psi_{out}\rangle = \exp\left(-i\hat{H}t\right) = e^{\alpha \hat{b}_{+}^{\dagger}/\sqrt{2}} \hat{S}_0 e^{\alpha \hat{b}_{-}^{\dagger}/\sqrt{2}} |0\rangle$$

$$i \frac{\partial \hat{\rho}_{HV}}{\partial t} = \left[\hat{H}; \hat{\rho}_{HV} \right]$$

$$\hat{\rho}_{HV} = e^{\alpha \hat{b}_H^{\dagger}} \hat{\rho}_V e^{\alpha^* \hat{b}_H}$$

$$i \frac{\partial \hat{\rho}_V}{\partial t} = \left[\hat{H}_V; \hat{\rho}_V \right]$$

$$\hat{H}_V = \Omega_R^H \left(\hat{d}_{\uparrow}^{\dagger} \hat{a}_{\uparrow} + \hat{d}_{\uparrow} \hat{a}_{\uparrow}^{\dagger} \right) + ig \left(\hat{b}_V \hat{d}_{\uparrow}^{\dagger} \hat{a}_{\uparrow} - \hat{b}_V^{\dagger} \hat{d}_{\uparrow} \hat{a}_{\uparrow}^{\dagger} \right)$$

$$g \ll \Omega_R^H$$

$$|\psi_{out}\rangle = \cos(\Omega_R t) \hat{a}_{\uparrow}^{\dagger} |0\rangle - i \sin(\Omega_R t) \hat{d}_{\uparrow}^{\dagger} |0\rangle + gt i \sin(\Omega_R t) \hat{a}_{\uparrow}^{\dagger} \hat{b}_V^{\dagger} |0\rangle - gt \cos(\Omega_R t) \hat{d}_{\uparrow}^{\dagger} \hat{b}_V^{\dagger} |0\rangle$$

$$\hat{d}_{\uparrow}^{\dagger}$$

$$\hat{b}_{-t}^{\dagger} \hat{a}_{\uparrow}^{\dagger}$$

$$|\psi_{out}\rangle = \cos(\Omega_R t) \hat{a}_{\uparrow}^{\dagger} |0\rangle - i \sin(\Omega_R t) \hat{b}_{\uparrow t}^{\dagger} \hat{a}_{\uparrow}^{\dagger} |0\rangle + gt i \sin(\Omega_R t) \hat{a}_{\uparrow}^{\dagger} \hat{b}_V^{\dagger} |0\rangle - gt \cos(\Omega_R t) \hat{b}_{\uparrow t}^{\dagger} \hat{a}_{\uparrow}^{\dagger} \hat{b}_V^{\dagger} |0\rangle$$

$$-i\hat{b}_{Vt}^\dagger\hat{a}_\uparrow^\dagger$$

$$|\psi_{out}\rangle = \cos(\Omega_R t)\hat{a}_\uparrow^\dagger|0\rangle - \sin(\Omega_R t)\hat{b}_{Vt}^\dagger\hat{a}_\uparrow^\dagger|0\rangle + gt\,i\sin(\Omega_R t)\hat{a}_\uparrow^\dagger\hat{b}_V^\dagger|0\rangle - igt\cos(\Omega_R t)\hat{b}_V^\dagger\hat{b}_{Vt}^\dagger\hat{a}_\uparrow^\dagger|0\rangle$$

$$|\psi_{out}\rangle = \cos(\Omega_R t)|0\rangle + (igt - 1)\sin(\Omega_R t)|1\rangle + igt\cos(\Omega_R t)|2\rangle$$

$$\hat{H}_+ = E_t\hat{d}_\uparrow^\dagger\hat{d}_\uparrow + \Omega_R\left(e^{-i\omega_L t}\hat{d}_\uparrow^\dagger\hat{a}_\uparrow + e^{-i\omega_L t}\hat{d}_\uparrow\hat{a}_\uparrow^\dagger\right) + g\left(\hat{b}_+\hat{d}_\uparrow^\dagger\hat{a}_\uparrow + \hat{b}_+^\dagger\hat{d}_\uparrow\hat{a}_\uparrow^\dagger\right) + E_t\hat{b}_+^\dagger\hat{b}_+$$

$$\hat{H}_+ = \Delta\hat{d}_\uparrow^\dagger\hat{d}_\uparrow + \Delta\hat{b}_+^\dagger\hat{b}_+ + \Omega_R\left(\hat{d}_\uparrow^\dagger\hat{a}_\uparrow + \hat{d}_\uparrow\hat{a}_\uparrow^\dagger\right) + g\left(\hat{b}_+\hat{d}_\uparrow^\dagger\hat{a}_\uparrow + \hat{b}_+^\dagger\hat{d}_\uparrow\hat{a}_\uparrow^\dagger\right)$$

$$\hat{H}_- = \Delta\hat{d}_\downarrow^\dagger\hat{d}_\downarrow + \Delta\hat{b}_-^\dagger\hat{b}_- + \Omega_R\left(\hat{d}_\downarrow^\dagger\hat{a}_\downarrow + \hat{d}_\downarrow\hat{a}_\downarrow^\dagger\right) + g\left(\hat{b}_-\hat{d}_\downarrow^\dagger\hat{a}_\downarrow + \hat{b}_-^\dagger\hat{d}_\downarrow\hat{a}_\downarrow^\dagger\right)$$

$$\hat{H}_B = \alpha B_x\left(\hat{a}_\downarrow^\dagger\hat{a}_\uparrow + \hat{a}_\uparrow^\dagger\hat{a}_\downarrow\right) + \alpha B_z\left(\hat{a}_\uparrow^\dagger\hat{a}_\uparrow - \hat{a}_\downarrow^\dagger\hat{a}_\downarrow\right) + \beta B_z\left(\hat{d}_\uparrow^\dagger\hat{d}_\uparrow - \hat{d}_\downarrow^\dagger\hat{d}_\downarrow\right)$$

$$\hat{a}_\uparrow^\dagger|0\rangle ; \hat{a}_\downarrow^\dagger|0\rangle ; \hat{d}_\uparrow^\dagger|0\rangle ; \hat{d}_\downarrow^\dagger|0\rangle ; \hat{b}_+^\dagger\hat{a}_\uparrow^\dagger|0\rangle ; \hat{b}_-^\dagger\hat{a}_\downarrow^\dagger|0\rangle ; \hat{b}_+^\dagger\hat{d}_\uparrow^\dagger|0\rangle ; \hat{b}_-^\dagger\hat{d}_\downarrow^\dagger|0\rangle$$

$$\hat{b}_-^\dagger\hat{a}_\uparrow^\dagger|0\rangle ; \hat{b}_+^\dagger\hat{a}_\downarrow^\dagger|0\rangle ; \hat{b}_-^\dagger\hat{d}_\uparrow^\dagger|0\rangle ; \hat{b}_+^\dagger\hat{d}_\downarrow^\dagger|0\rangle$$

$$|\uparrow\rangle + |\downarrow\rangle \rightarrow |\uparrow R\rangle + |\downarrow L\rangle$$

$$|\uparrow\rangle + |\downarrow\rangle \rightarrow (|\uparrow\rangle + |\downarrow\rangle)|R\rangle + (-|\uparrow\rangle + |\downarrow\rangle)|L\rangle$$

$$|\uparrow\rangle - |\downarrow\rangle \rightarrow | +_1 +_2 \rangle - | -_1 -_2 \rangle$$

$$\Omega_R^H \sim g\sqrt{N}$$

$$\frac{\partial \hat{\rho}}{\partial t} = -i\left[\hat{H};\hat{\rho}\right] - \hat{L}(\hat{\rho})$$

$$\hat{L}(\hat{\rho}) = \gamma_{PD} \left(\hat{\rho} - \hat{\sigma}_z \hat{\rho} \hat{\sigma}_z \right)$$

$$\hat{\rho}_{in}$$

$$\hat{\rho}_{out} = \hat{S}_\tau(\hat{\rho}_{in})$$

$$\hat{\rho}_{ph} = \hat{M}_{spin\; relax}(\hat{\rho}_{out})$$

$$= f_{+x}(B/T)\langle x|\hat{\rho}_{out}|x\rangle + f_{-x}(B/T)\langle -x|\hat{\rho}_{out}|-x\rangle$$

$$f_{+x}(B/T) = \frac{e^{B/T}}{e^{B/T} + e^{-B/T}}$$

$$f_{-x}(B/T) = \frac{e^{-B/T}}{e^{B/T} + e^{-B/T}}$$

$$= f_{+x}(B/T)|x\rangle\langle x| + f_{-x}(B/T)|-x\rangle\langle -x|$$

$$\sqrt{P}$$

$$|000\rangle + |001\rangle + |010\rangle - |011\rangle + |100\rangle + |101\rangle - |110\rangle + |111\rangle$$

$$F = (\langle +_1 +_2 | + \langle -_1 -_2 |) \hat{\rho}_{out} (| +_1 +_2 \rangle + | -_1 -_2 \rangle)$$

$$\hat{\rho}_{ph} = \hat{\rho}_0 + \hat{\rho}_1 + \hat{\rho}_2$$

$$p_1 = Tr\left(\hat{\rho}_1\right) \quad ; \quad p_2 = Tr\left(\hat{\rho}_2\right)$$

$$\hat{\rho}_2^{(2\pi)} \approx (g\tau)^2 \left(\frac{\Omega_R^2}{\Omega_R^2 + \Delta^2} \right)^2 \begin{pmatrix} 1 & 0 & 0 & \tanh^2(B/T) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \tanh^2(B/T) & 0 & 0 & 1 \end{pmatrix}$$

$$g^{(2)}(0)=\frac{2\left|\delta\right|^2\cos^2(\theta/2)}{\left(\sin^2(\theta/2)+\left|\delta\right|^2\right)^2}$$

$$|0\rangle=|H\rangle$$

$$|1\rangle=|V\rangle$$

$$\phi_{ab}^+=\frac{1}{\sqrt{2}}\left(H_aH_b+V_aV_b\right)$$

$$\frac{\mu g B}{T}$$